

## Exam: One-Way ANOVA

*Multiple Choice (Single Select) - 4 Answer Options*

### 1. When do you use ANOVA?

- A) Comparing 3+ groups
- B) Relationship between two metric variables
- C) Comparing 2 groups
- D) Normality test

### 2. Which variance is in the numerator of the F-statistic?

- A) Error variance
- B) Variance within groups
- C) Total variance
- D) Variance between groups

### 3. What does a significant F-test mean?

- A) All means differ
- B) Variances are equal
- C) Samples are representative
- D) At least one mean parameter differs

### 4. Which post-hoc test is the most conservative?

- A) LSD
- B) Scheffe
- C) Bonferroni
- D) Tukey HSD

### 5. What is the effect size measure in ANOVA?

- A) Cramer's V
- B) Cohen's d
- C) Eta-squared
- D) R-squared

**6. Which assumption is central to ANOVA?**

- A) Ordinal DV
- B) Metric IV
- C) Equal sample sizes
- D) Homogeneity of variances

**7. What is the purpose of post-hoc tests?**

- A) Testing variances
- B) Locating significant differences
- C) Normality testing
- D) Power analysis

**8. What does omnibus test mean?**

- A) Test of individual differences
- B) Test of variance homogeneity
- C) Overall difference test
- D) Interaction test

## Solutions

1: A

2: D

3: D

4: B

5: C

6: D

7: B

8: C